

— FINAL PROJECT

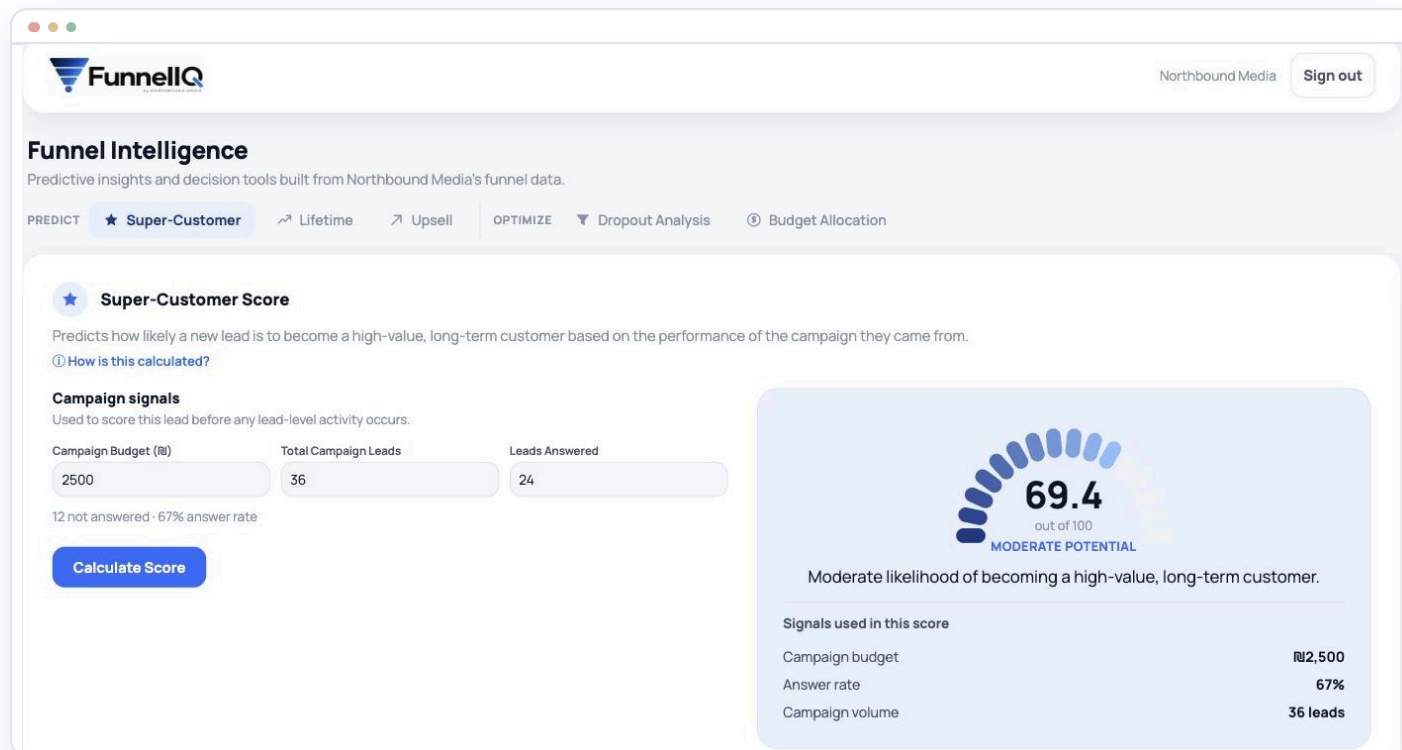
FunnelIQ

From 3,500 historical funnel records to a deployed prediction and optimization tool.

Yasmin Torgeman

A deployed workspace for prediction and optimization

Five decision tools built on Northbound Media's campaign data.



PREDICT

Super-Customer · Lifetime · Upsell

OPTIMIZE

Follow-up Dropout · Budget Allocation

5 DECISION TOOLS

AUTHENTICATED

DEPLOYED

One dataset, five business questions

3,500 raw records → validation & cleaning → 3,490 analysis-ready records

PREDICT

- How long will a customer stay?
- Who is likely to upsell?
- Which campaigns attract high-value customers?

OPTIMIZE

- When should sales stop following up?
- How should ad budget be allocated?

— PROJECT PROCESS

From raw data to a deployed tool



EARLY SIGNAL Mid-budget campaigns (€2,000–5,000) had higher conversion rates than higher-spend campaigns. This pattern re-emerges across other analyses.

Leakage: a question of timing, not correlation

Only use information available at prediction time.

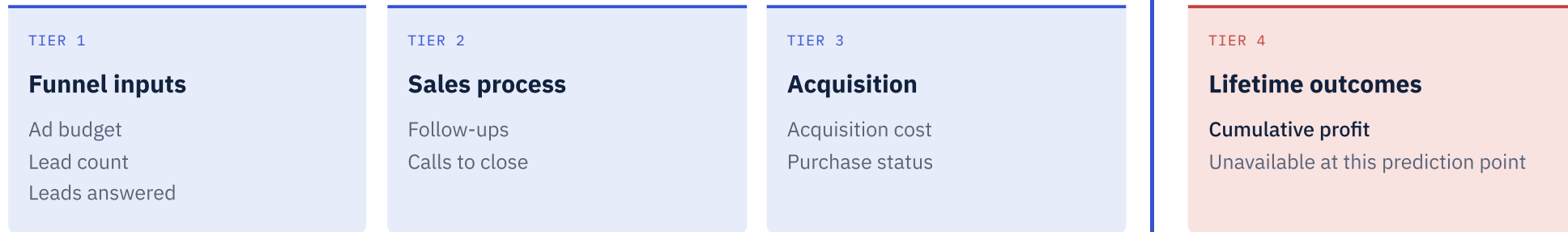
EXAMPLE: PREDICTING CUSTOMER LIFETIME

Available by acquisition

LTV prediction point

At / just after acquisition

Future information



For the LTV model, tiers 1–3 are eligible inputs. Other lifetime outcomes are excluded.

DECISION

Excluded `cumulative_profit` despite its strong correlation with LTV ($r = 0.85$): it contains lifetime information unavailable at prediction time.

From features to business answers

LIFETIME

How long will this customer stay?

PREDICTIVE PERFORMANCE

2.87 months RMSE

vs. 12.43 months for the naive baseline

Key signal
Calls to close

BUSINESS USE

More calls to close flags customers for stronger onboarding.

UPSELL

Who should be prioritized for upsell?

PREDICTIVE PERFORMANCE

0.785 ROC-AUC

Ranks customers by upsell likelihood

Key signals
Calls to close and acquisition cost

MODEL VS. SIMPLE RULE

Model recall: **83%**, versus **59%** for the simple rule.

SUPER-CUSTOMER

Which campaigns attract high-value customers?

MAIN DRIVER

**Campaign
budget tier**

~75% of model feature importance

Pattern
Mid-budget campaigns score highest

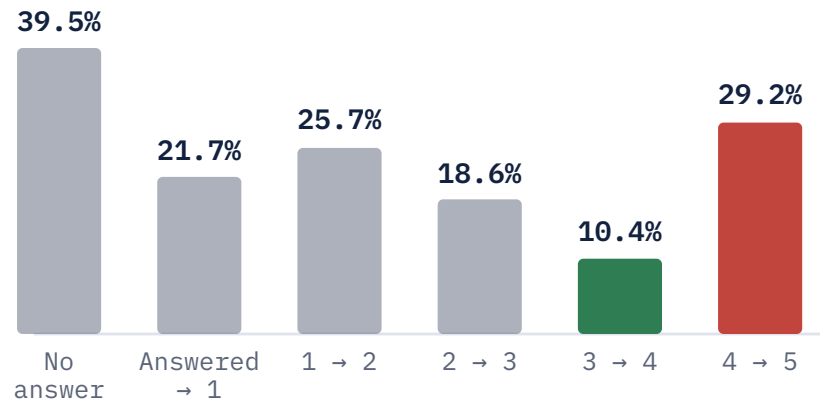
SCOPE

Scores the campaign a lead came from, rather than individual lead characteristics.

Reconsider the follow-up cutoff

Lowest dropout: between rounds 3 and 4.

OBSERVED DROPOUT RATE BY TRANSITION



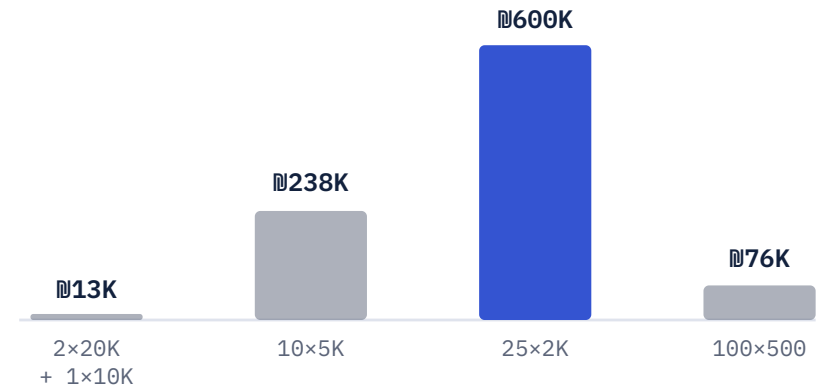
Investigate the jump entering round 5 (29.2%) before changing the sequence. Dropout between rounds 3 and 4 is only **10.4%**.

All recorded closes follow round 5. This dataset cannot establish the effect of stopping earlier.

Same ₪50K. Different allocations.

Compare predicted profit for the same spend.

MODEL-SIMULATED PROFIT • NOT REALIZED PROFIT



25 campaigns × ₪2,000 produced the highest simulated profit among the tested allocations. **Validate with a pilot.**

~46× the predicted profit of the most concentrated option. Labels show campaign count × budget (₪).

Two patterns recur across the analyses

Fewer calls to close

ASSOCIATED WITH

Longer customer lifetime

Higher upsell likelihood

Higher profit

Mid-budget campaigns

€2,000–5,000 per campaign

ASSOCIATED WITH

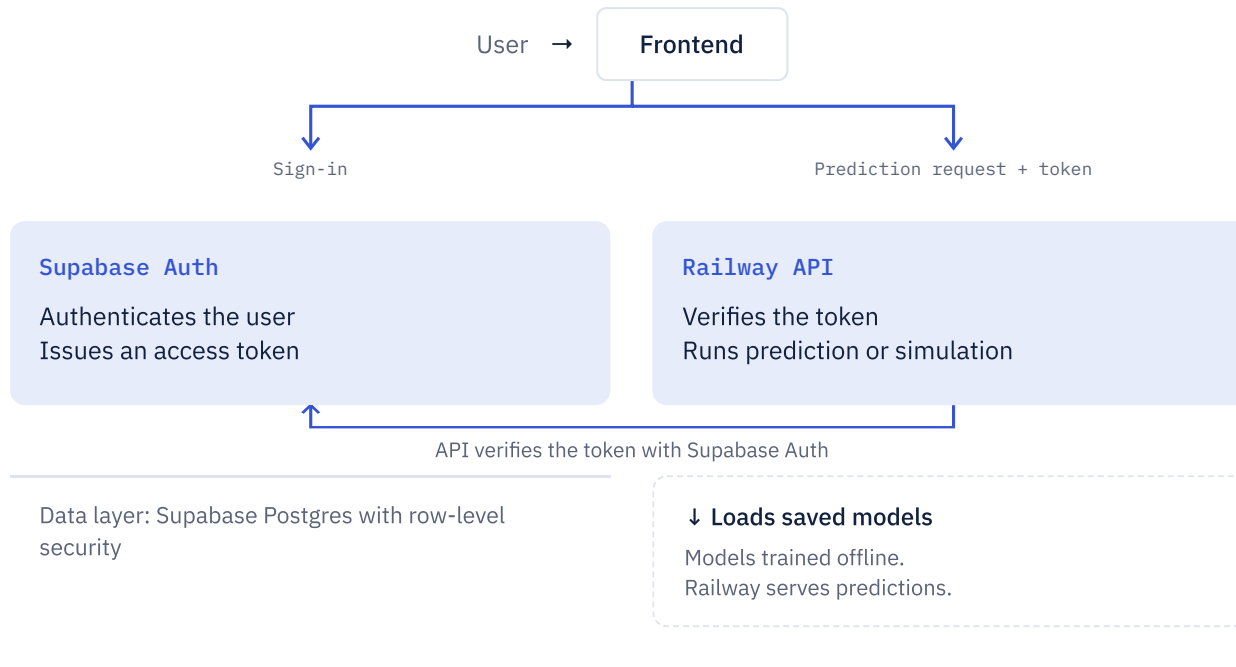
Higher conversion rates

Higher predicted referral likelihood

The patterns recur across models and analyses of the **same dataset**. They still need validation on new data.

These are observational associations, not evidence of causality.

From trained models to a deployed application



Authenticated requests

The API verifies the user's token before prediction or simulation.

4 model endpoints

3 prediction endpoints
1 budget simulation endpoint
Follow-up analysis is the fifth UI tool.

Pull requests and automated checks

Syntax checks and an API import check that loads all saved models.

GITHUB WORKFLOW

Branch → pull request → automated checks → merge

From prediction to action

Next: use model predictions and customer context to suggest an action.

Planned · Not yet implemented



ILLUSTRATIVE EXAMPLE

Model prediction

Predicted lifetime: **8 months**

Customer context

Many calls to close

Suggested action

Prioritize onboarding and schedule an early retention check-in.

➤ [GitHub](#)

➤ [Live App](#)

➤ [Project report](#)